

Education

IIT (BHU), Varanasi

CGPA – 8.26

Bachelor of Technology

2019 – 2023

Experience

Goldman Sachs

Jun. 2025 – Present

Analyst

- **Banker Copilot:** Developed a full-stack agentic AI platform using **React, TypeScript, Python, FastAPI, LangGraph, LangChain, AWS Bedrock, RAG, PostgreSQL/pgvector, Redis, and Celery**, enabling multi-agent financial research, secure semantic retrieval, peer analysis, and cited report generation with asynchronous document ingestion and research execution.
- **Equity Derivatives:** Developed **ML-driven options trading analytics** using **C++, Golang, Python, Kafka, Redis, and XGBoost**, integrating real-time market data, **volatility forecasting**, Greeks-based risk calculations, and historical strategy simulation; deployed containerized services using **Docker and Kubernetes**.

Casahealth Tech Pvt Ltd

Feb. 2025 – Jun. 2025

Software Engineer II

- **Healthcare Workflow Engine:** Built a multi-tenant EMR platform using **Java, Spring Boot, RabbitMQ, Redis, PostgreSQL, and Docker**, automating appointment scheduling, clinical workflows, and patient record management through event-driven processing, configurable state transitions, idempotent execution, retries, and audit logging.

Edfora Infotech Pvt. Ltd.

Jun. 2023 – Feb. 2025

Software Engineer

- **Assessment Engine:** Built a scalable online/offline assessment platform using **C++, Rust, Node.js, Kafka, Redis, MongoDB, and AWS SQS**, supporting **100K+ concurrent users** and processing **80K+ exam responses**; optimized scoring, rankings, and leaderboards, reducing latency by **40%** and improving throughput by **35%**.
- **AI Knowledge Engine:** Built a semantic question-processing platform using **Python, FastAPI, LangChain, Sentence Transformers, FAISS/Pinecone, MongoDB, and AWS SQS**, integrating OCR/LaTeX extraction, NLP classification, and vector similarity search across **2M+ questions**; reduced duplicates by **95%**.
- **Personalized Learning Engine:** Developed an ML-driven recommendation system using **Python, FastAPI, XGBoost, Scikit-learn, spaCy, and React**, combining student performance analytics and semantic question metadata to identify learning gaps and recommend personalized practice; improved engagement by **20%**.

Projects

- **Quant Trading & Research Platform** | [GitHub](#) | *In Progress*
Developing an algorithmic trading platform using **Python, Pandas, DuckDB, Parquet, WebSockets, XGBoost, Optuna, and Streamlit**, integrating OHLCV and bid-ask data, event-driven backtesting, walk-forward ML optimization, pre-trade risk controls, and Telegram alerts.
- **VisionAI** | [Live Demo](#) | [GitHub](#) | *React, Vite, Redux Toolkit, Tailwind CSS, Monaco Editor, Node.js, Express.js, LangChain, LangGraph, Qdrant, Gemini, Groq, OpenRouter, Tavily, Redis, MongoDB, Firebase, Razorpay, Docker, AWS ECS, ECR, S3, CloudFront, GitHub Actions*
- **AI HireBooster** | [Live Demo](#) | *React, Vite, Tailwind CSS, Redux Toolkit, Recharts, jsPDF, Web Speech API, Node.js, Express.js, MongoDB, Mongoose, JWT, Firebase Auth, Multer, pdfjs-dist, OpenRouter, GPT-4o-mini, Razorpay*

Technical Skills

- **Languages:** C++, Rust, Python, Golang, Java, JavaScript, TypeScript, SQL
- **Frameworks & Libraries:** FastAPI, Spring Boot, Node.js, Express.js, React, Angular, GraphQL, NumPy, Pandas
- **Databases, Search & Observability:** PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, OpenSearch, Pinecone, FAISS, pgvector, DuckDB, Prometheus
- **Cloud & DevOps:** AWS, AWS Bedrock, SQS, Kafka, RabbitMQ, Docker, Kubernetes, Jenkins, Git, Linux, Parquet
- **Machine Learning:** Scikit-learn, XGBoost, PyTorch, Feature Engineering, Model Evaluation, Time-Series Analysis, Statistical Modeling, Volatility Forecasting
- **AI & NLP:** RAG, Agentic AI, LangChain, LangGraph, LLMs, Sentence Transformers, Semantic Search, Vector Embeddings, OCR
- **Core & Domain:** Data Structures, Algorithms, OOP, DBMS, REST APIs, Microservices, Distributed Systems, Multithreading, Concurrency, Low-Latency Computing, Backtesting, Options Analytics, Greeks, Risk Modeling, P&L Attribution

Achievements

- Secured **Global Rank 402** among **100K participants** in **TCS CodeVita Season 10**.
- **Finalist, Jumpstart '22** by Publicis Sapient; ranked in the **top 5%** among 29K+ participants.